COURSE CODE
SBT-DF202

COURSE TITLE
Computer and Digital Forensics

LAB TITLE
Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis

NAME
Kumpon Emmanuel Nandom

STUDENT ID
2025/FWSD/11325

DATE
29/08/2026

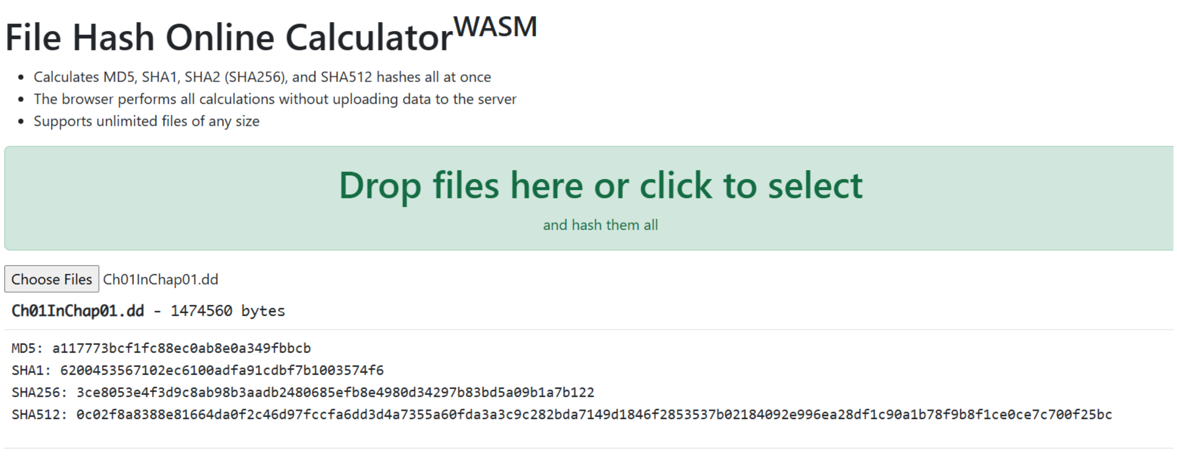
PLATFORM USED
Window Autopsy Software and Kali Linux

INSTRUCTOR
Mr. Aminu Idris, AMCPN
Commandant

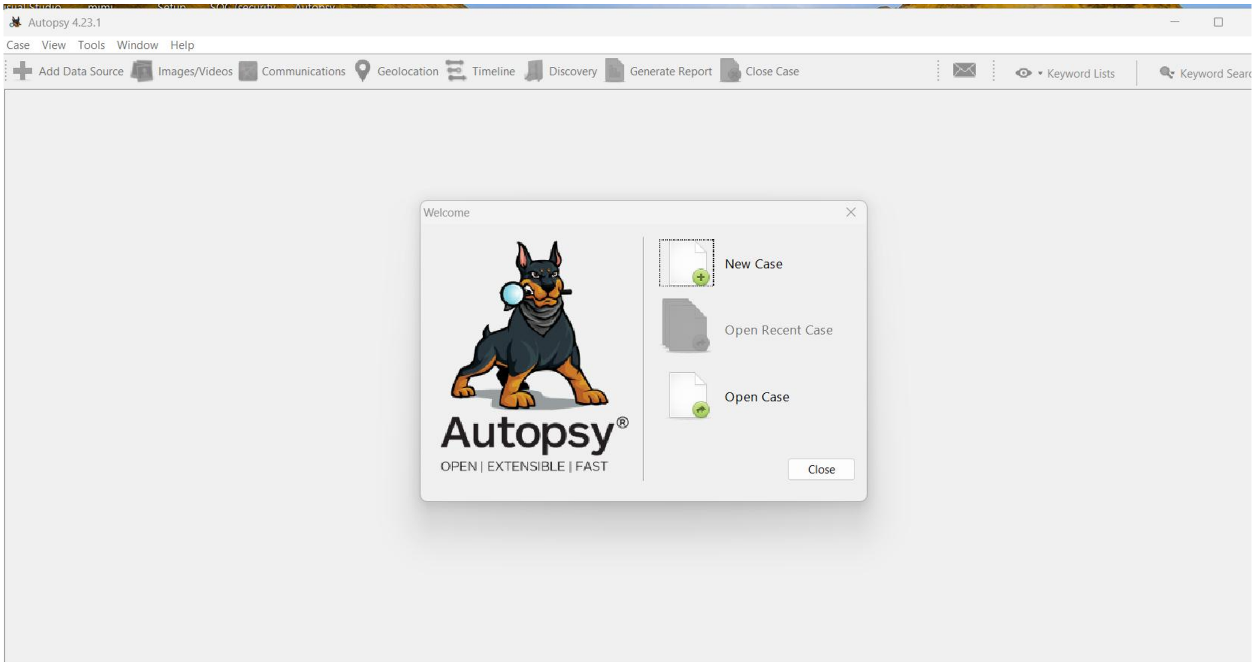
Objective

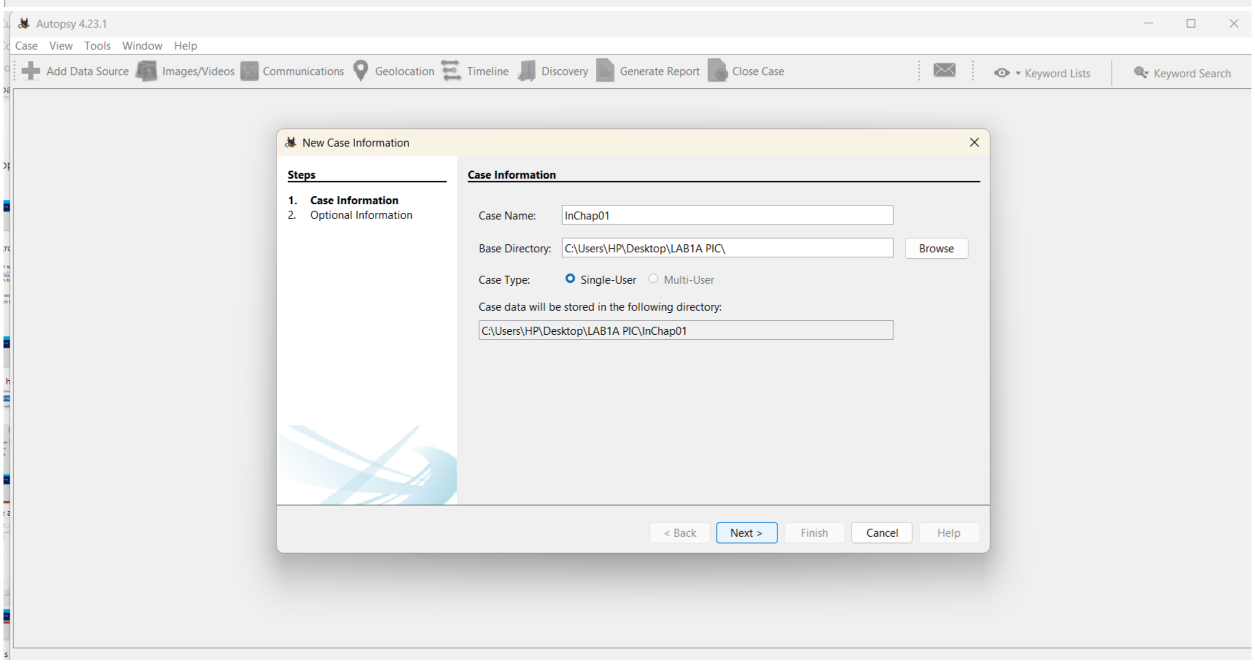
By completing this lab, trainees will demonstrate their ability to apply the digital forensic investigation process in a controlled practical environment. Students will preserve and hash a forensic disk image, perform structured analysis using Autopsy, examine the image using The Sleuth Kit command-line tools, identify and recover deleted files, analyse file metadata and data units, perform keyword searches, verify recovered evidence using cryptographic hashes, and prepare a professional forensic report containing repeatable findings.

Check hash code online

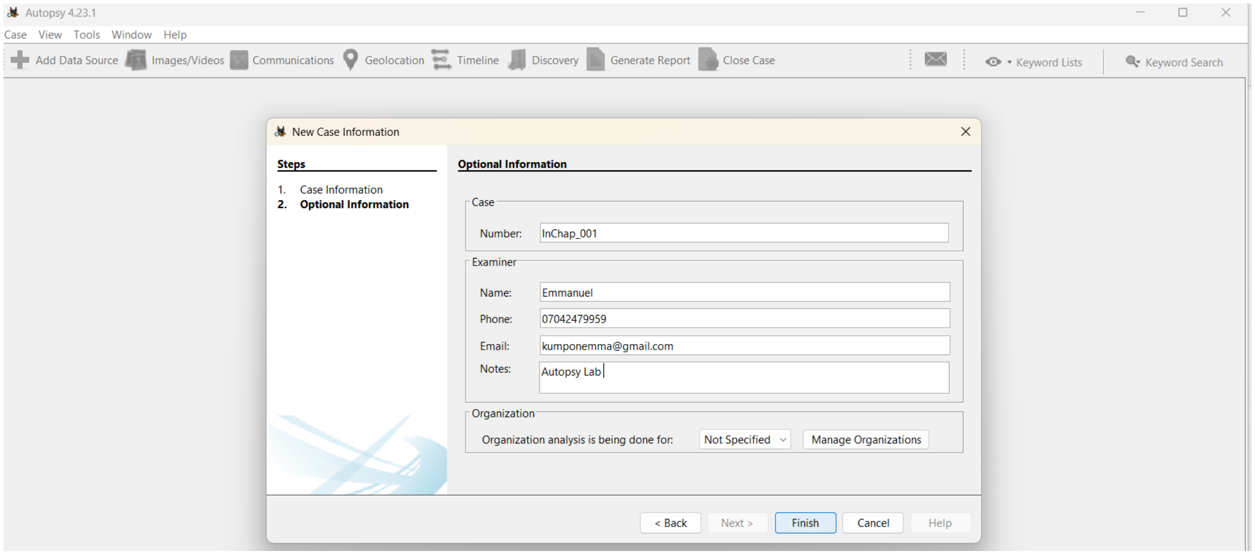


Create a case with name

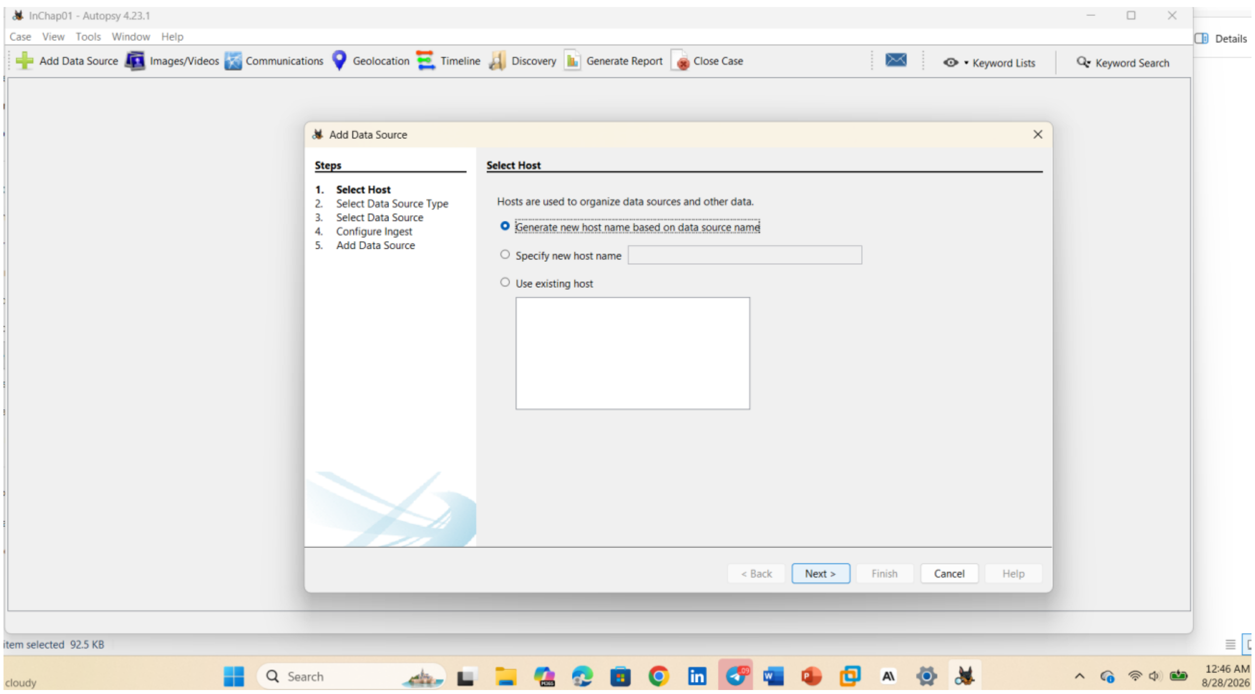




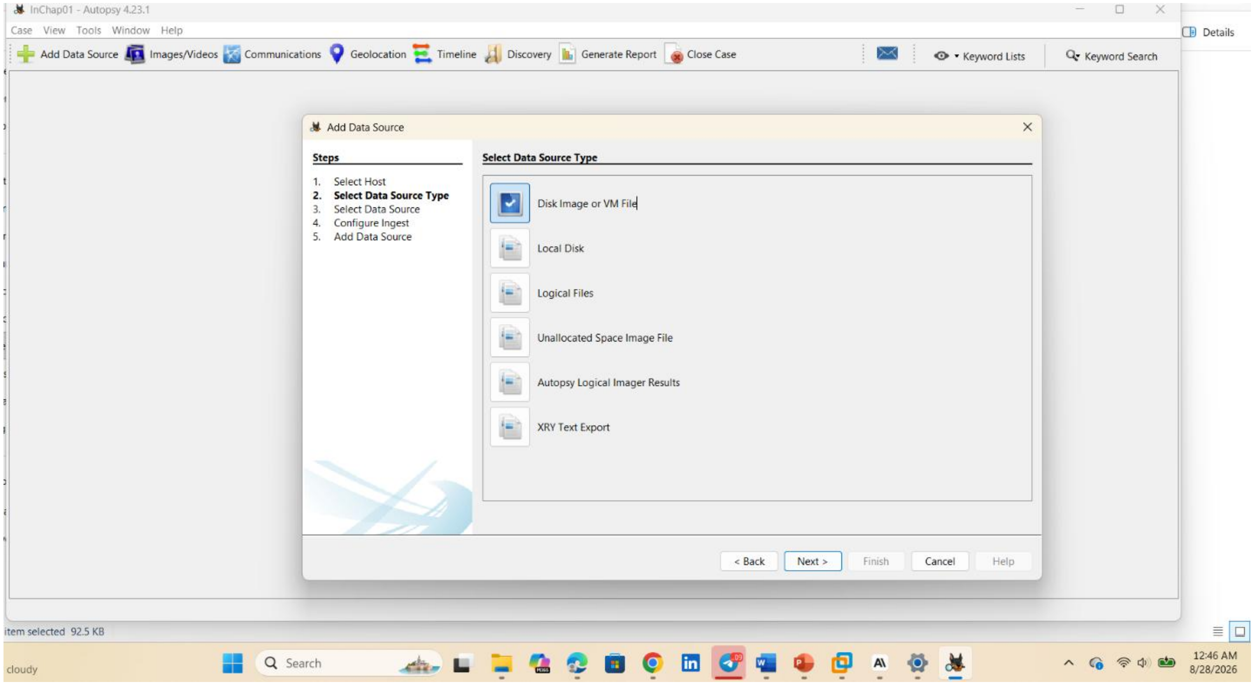
Details of the case



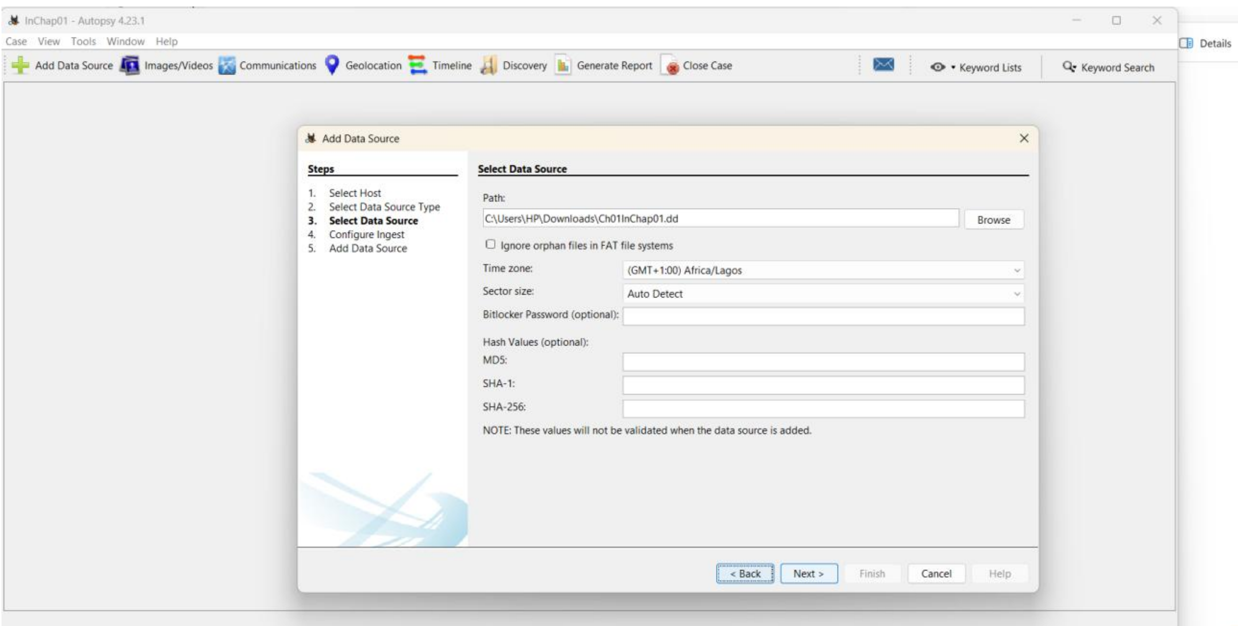
Choose Date Format



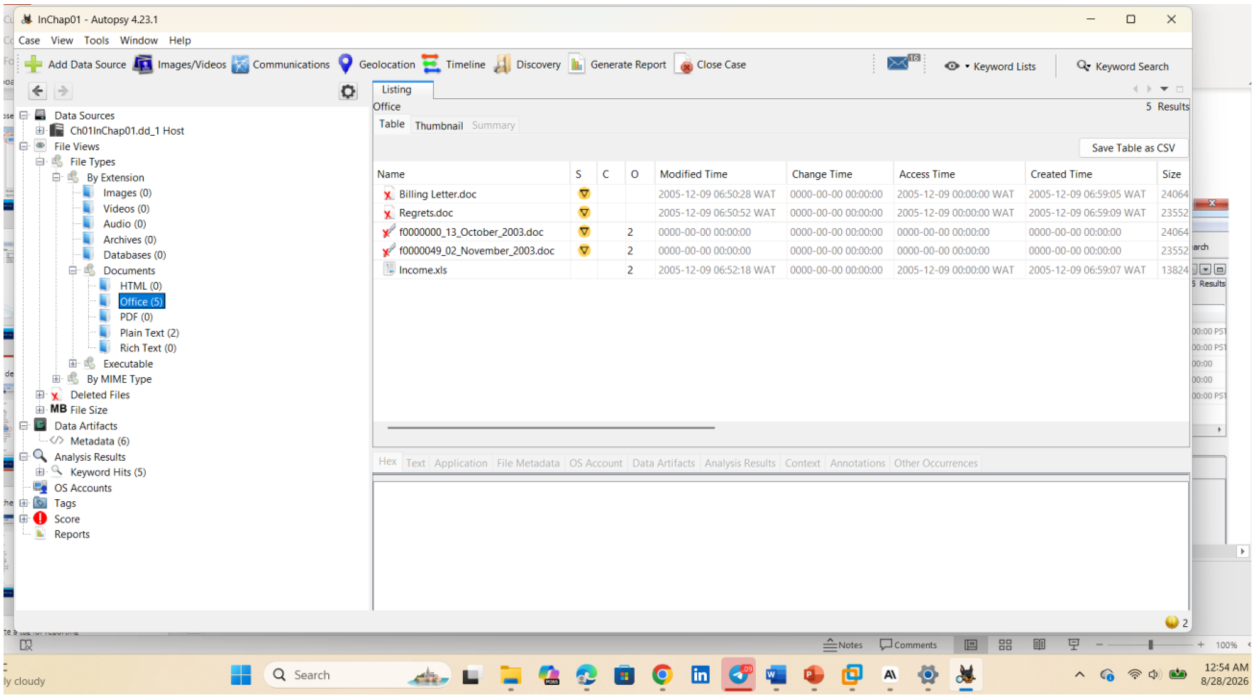
Click Disk Image or VM File



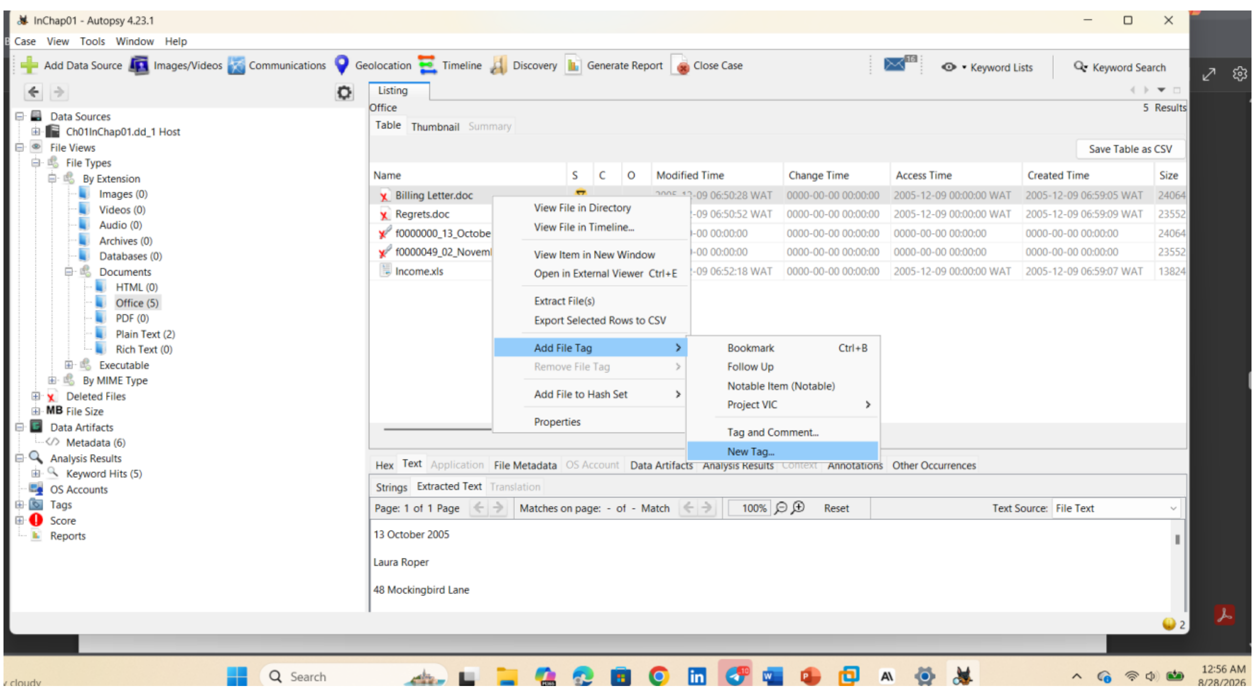
Choose the image file



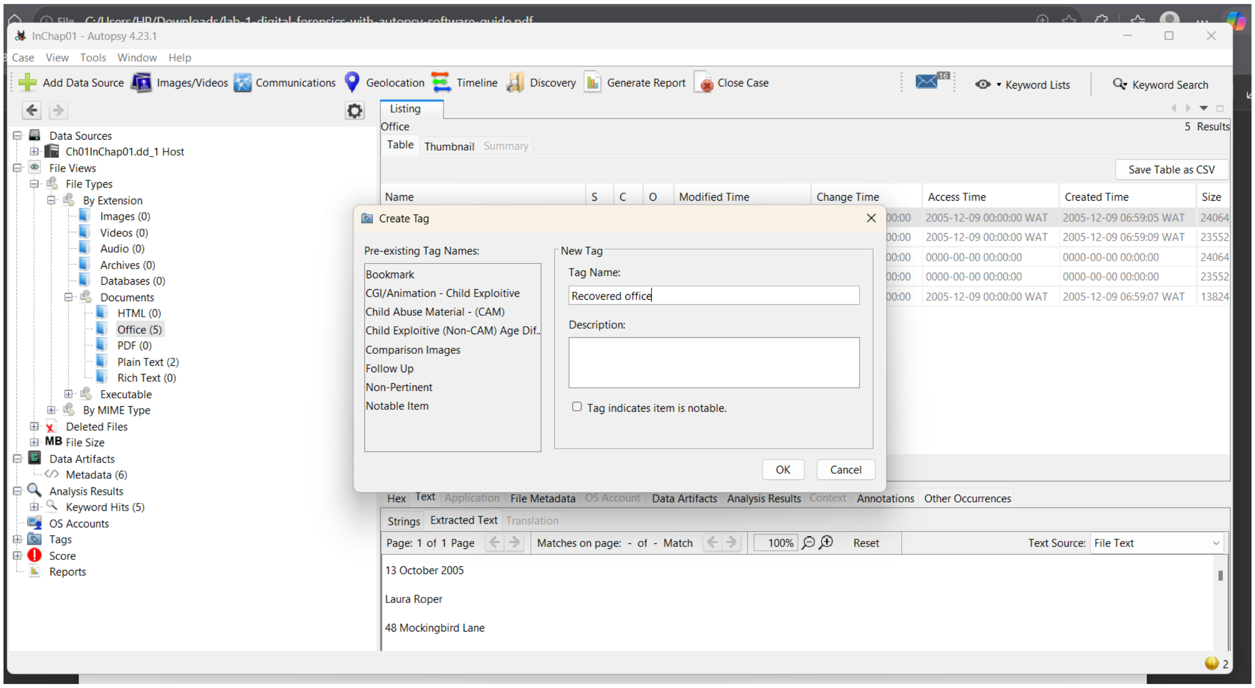
Find deleted files



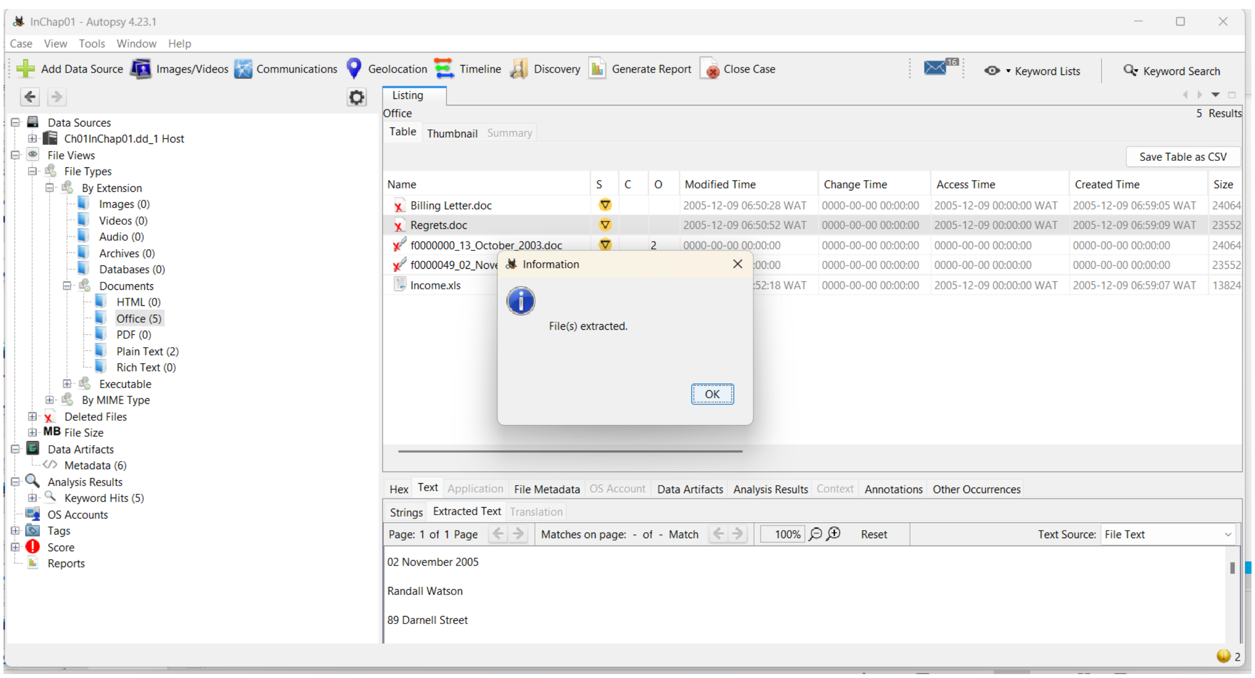
Tag the file



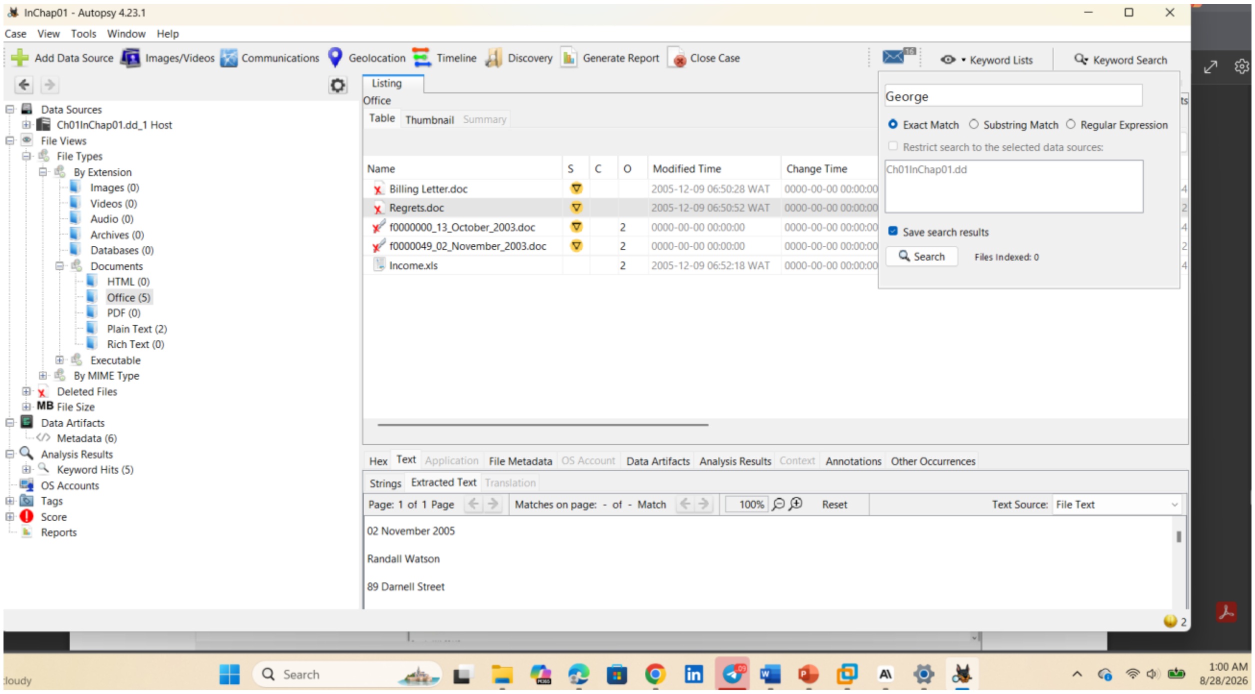
Create a tag reporting

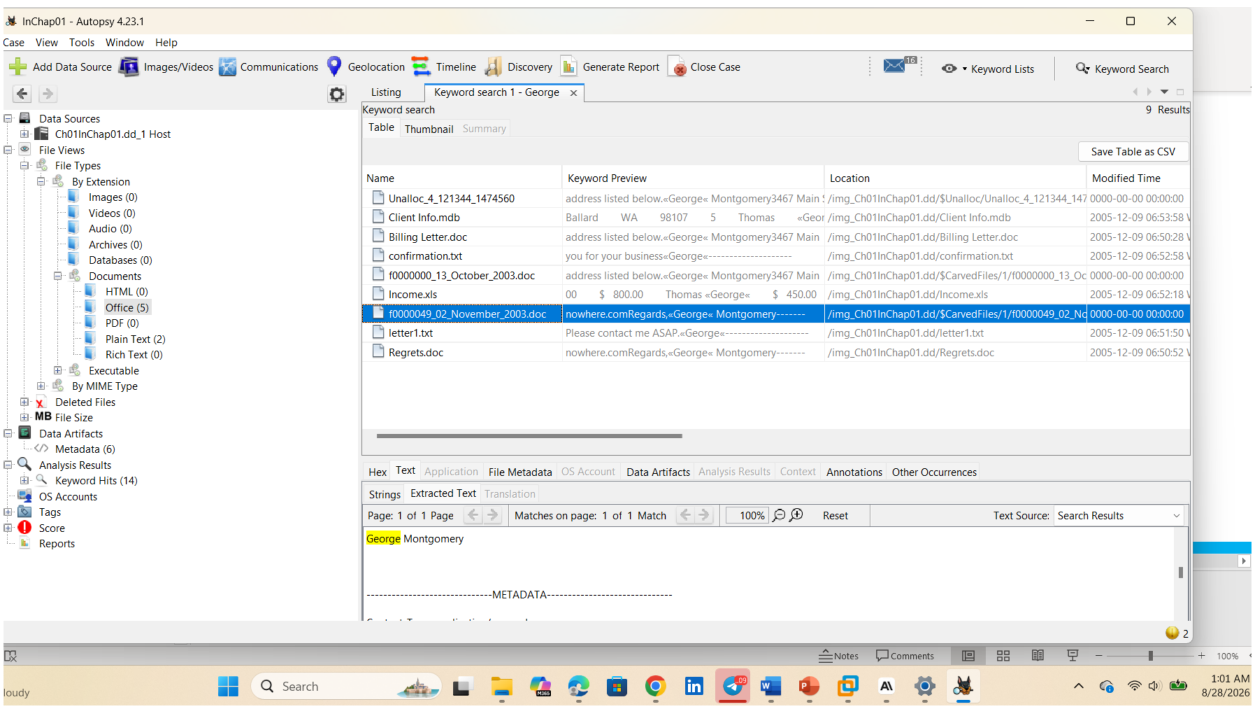


Recover deleted file

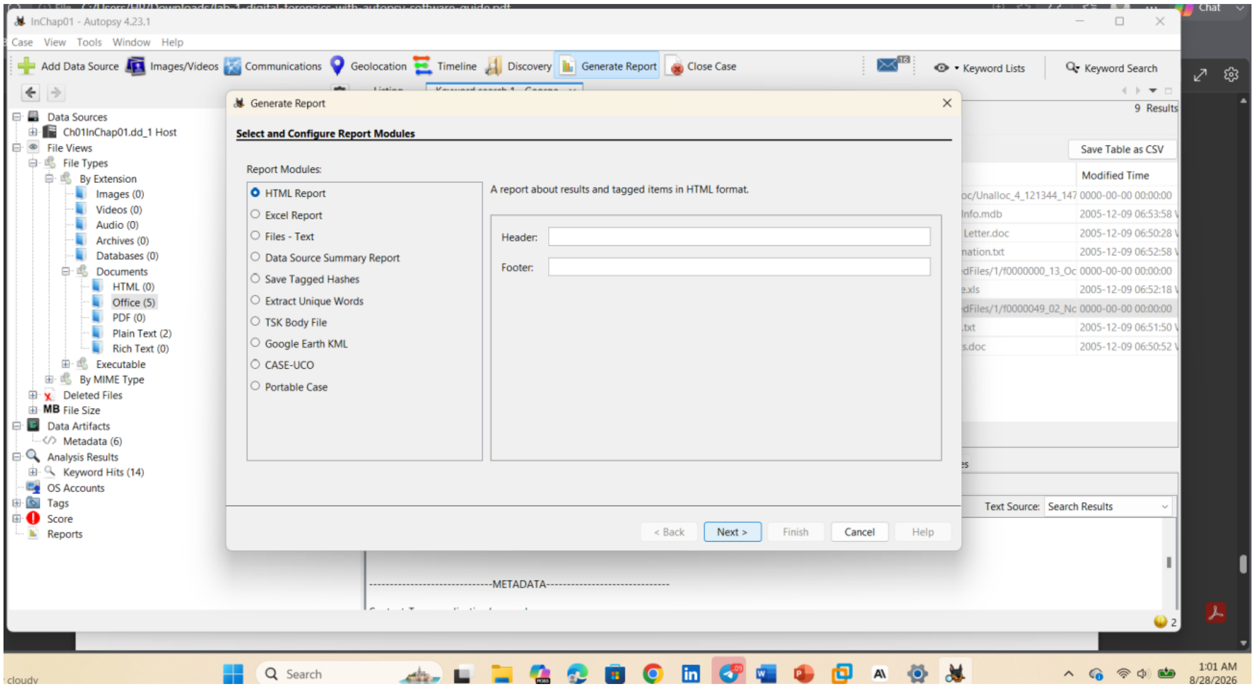


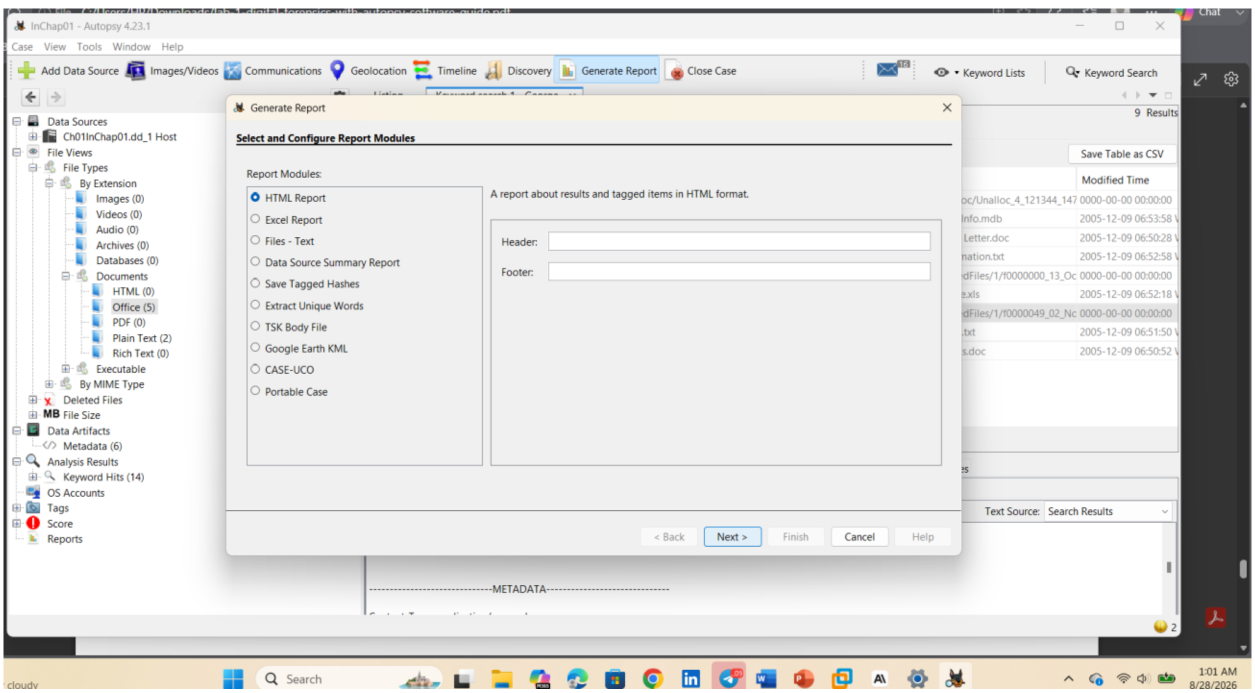
Search keywords “George”

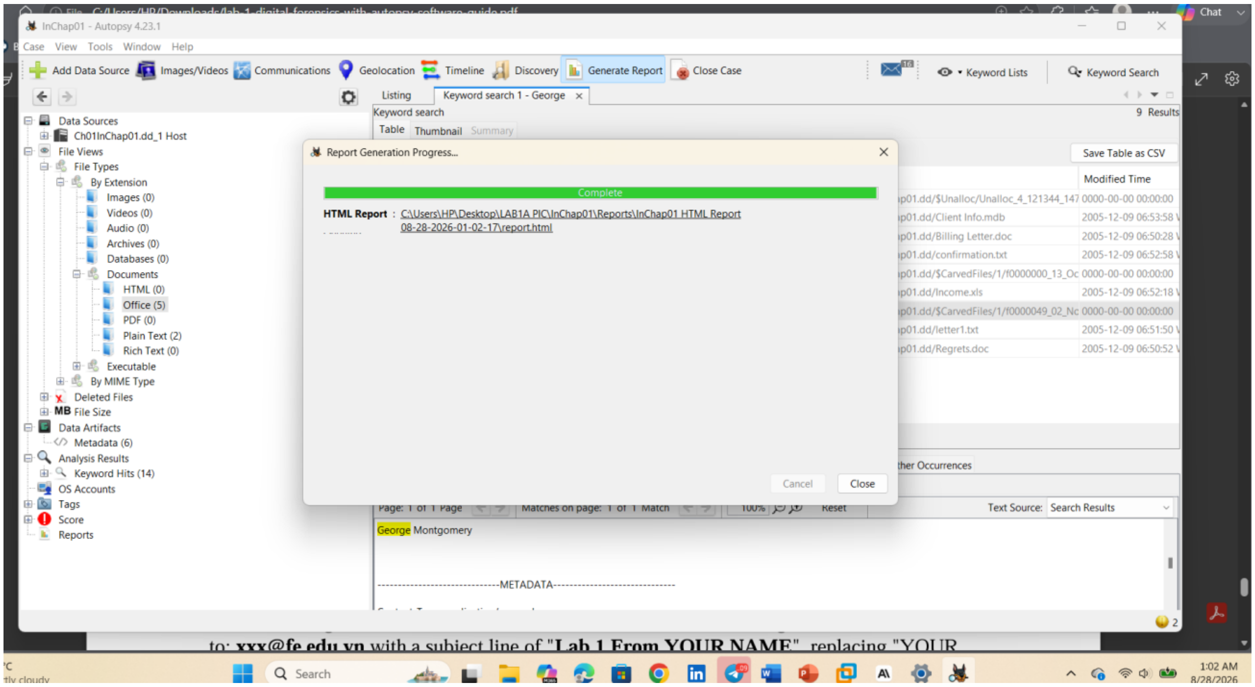




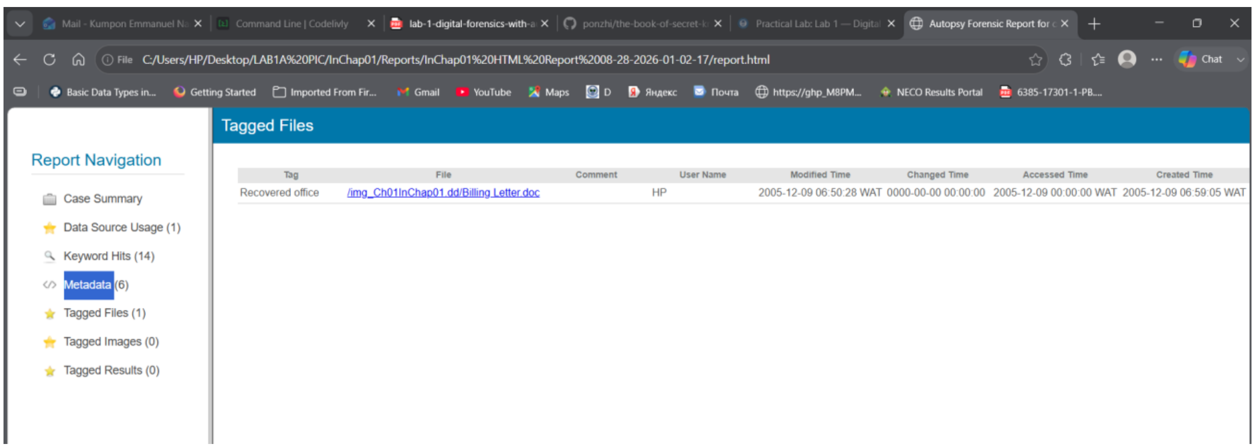
Generate reports





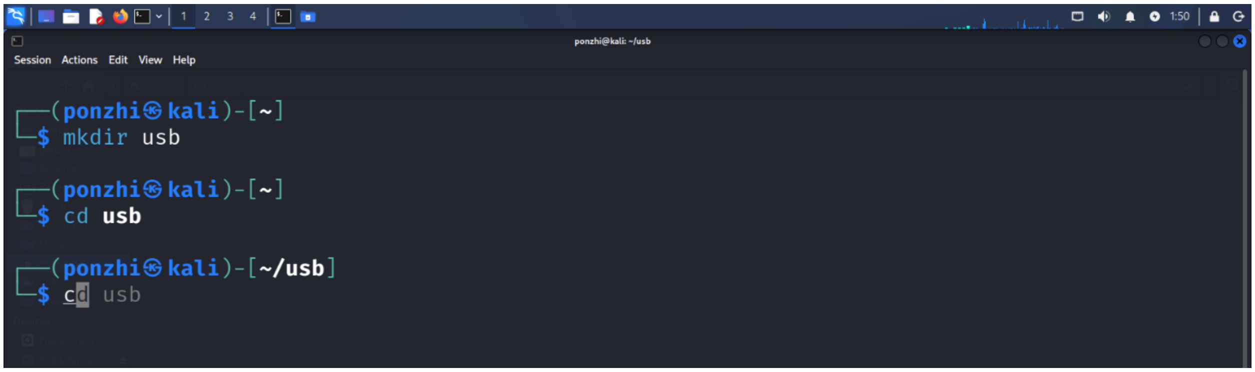


Open or save as file doc recovered

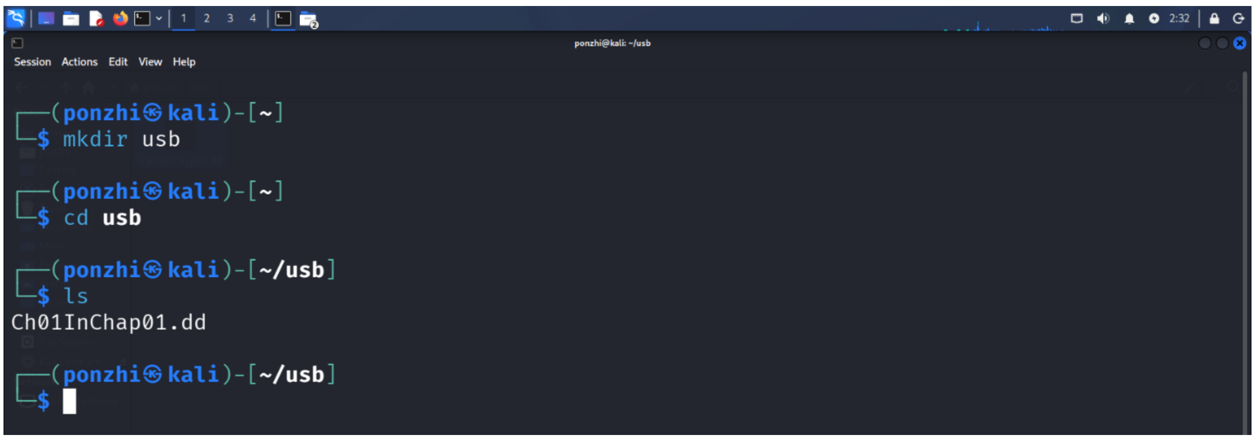


Part B Sleuth Kit Analysis

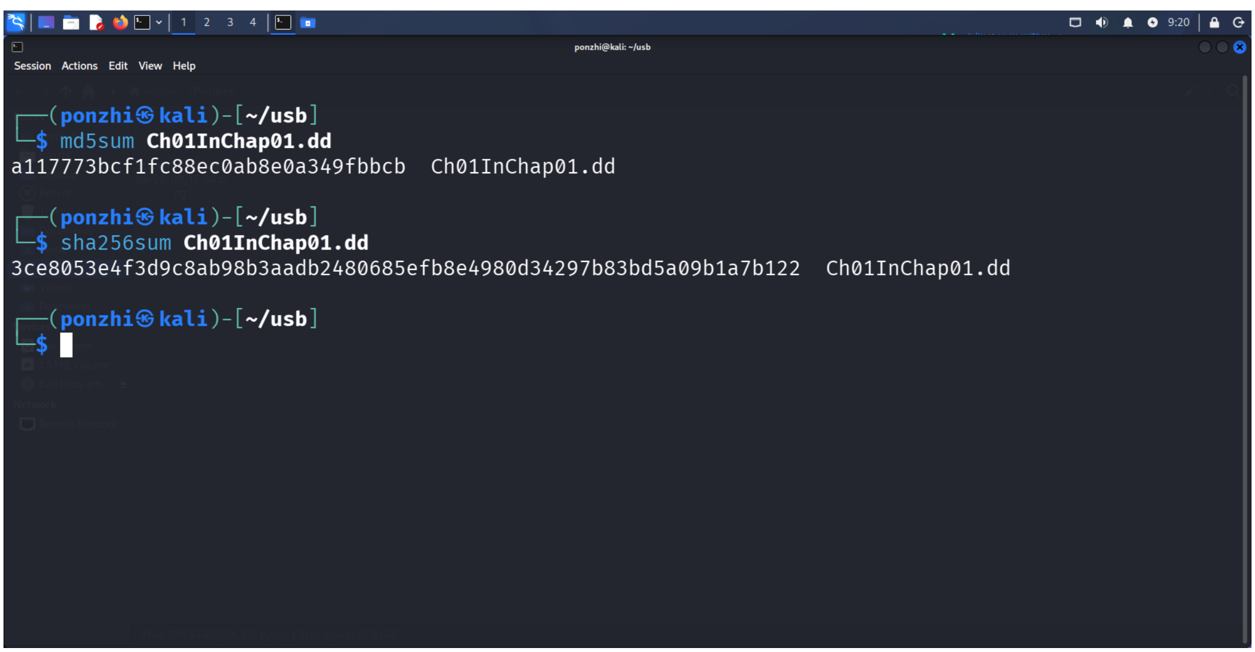
Dedicated Lab folder



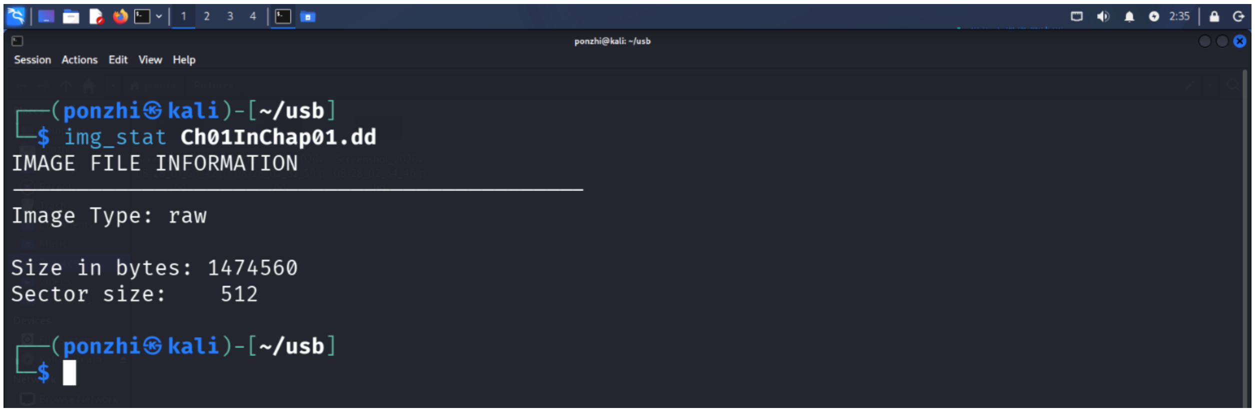
Moving the Forensic image into Usb



MD5 and SHA-256 hashes of Ch01InChap01

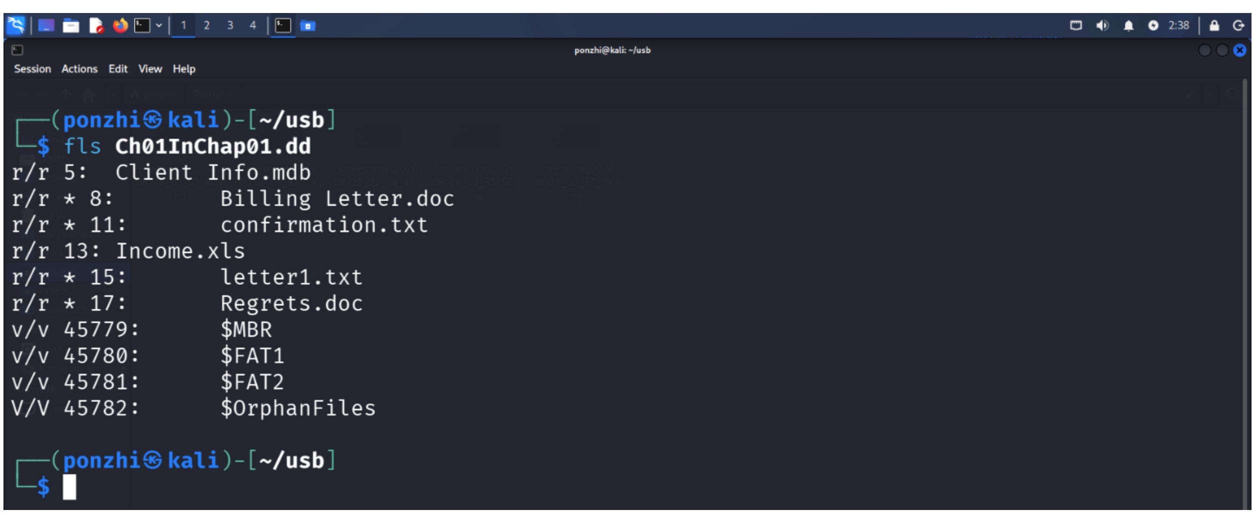


Showing the details of the image format

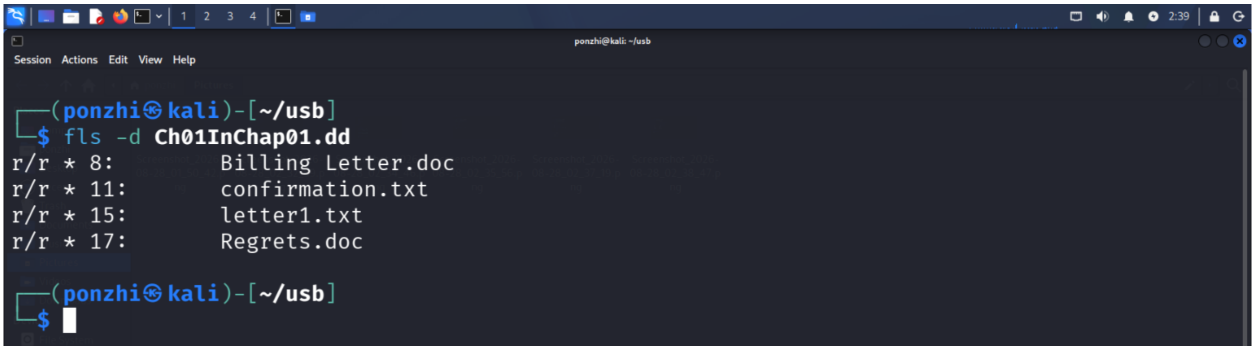


Showing file system details and statistics, including layout, sizes and labels

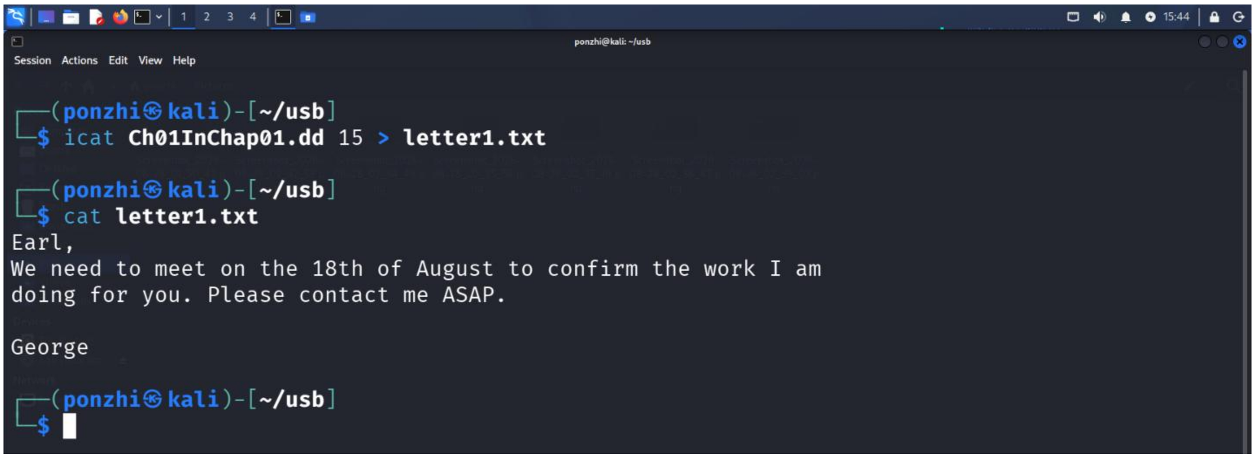
List allocated and deleted file names in a directory



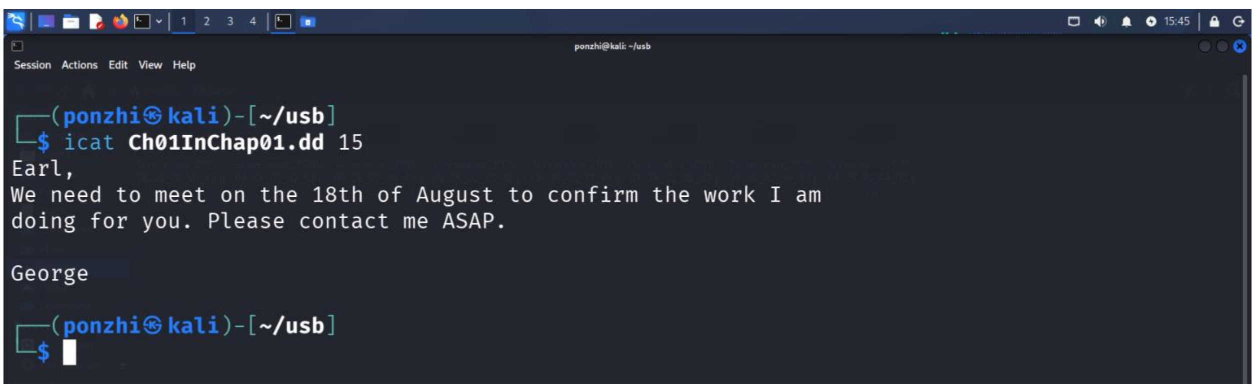
-d Display deleted entries only



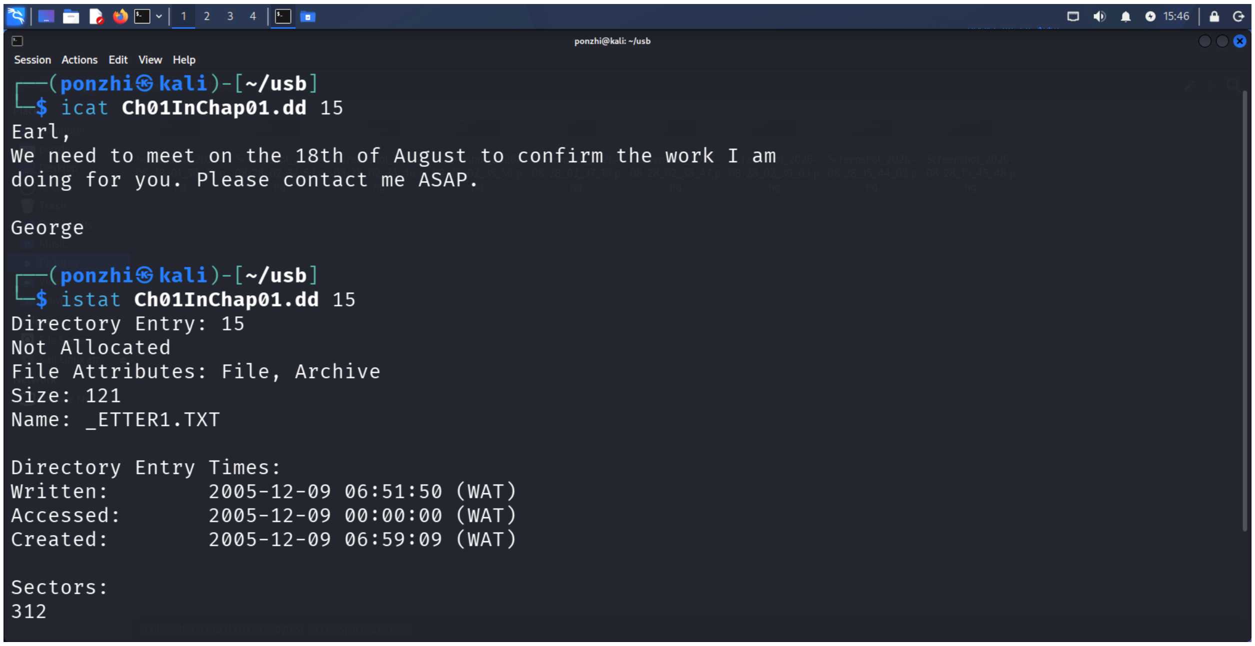
Recover deleted files using inode without specifying a file name



Extract the data units of letter1.txt using inode 15

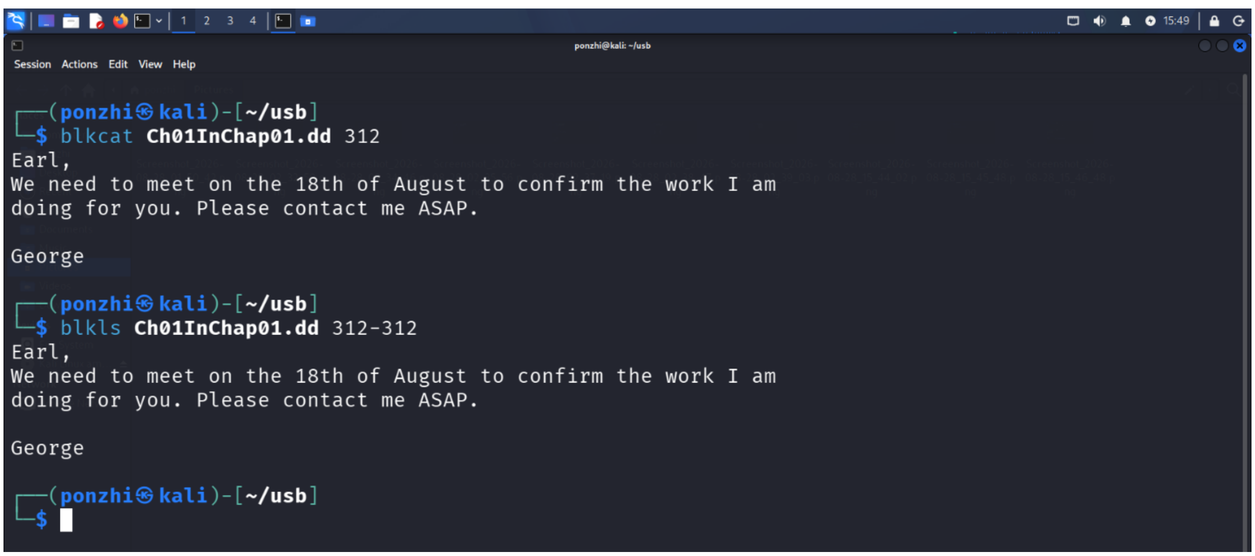


Display the statistics and details about a given meta data structure

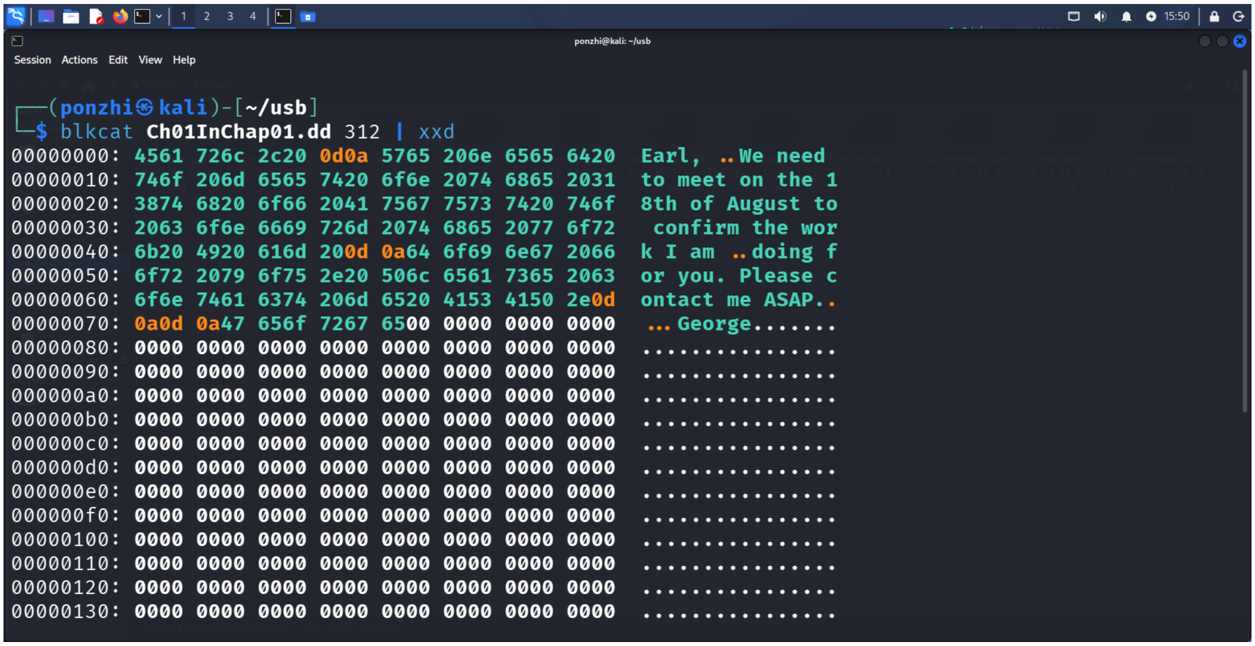


Extract the contents of a given data unit.

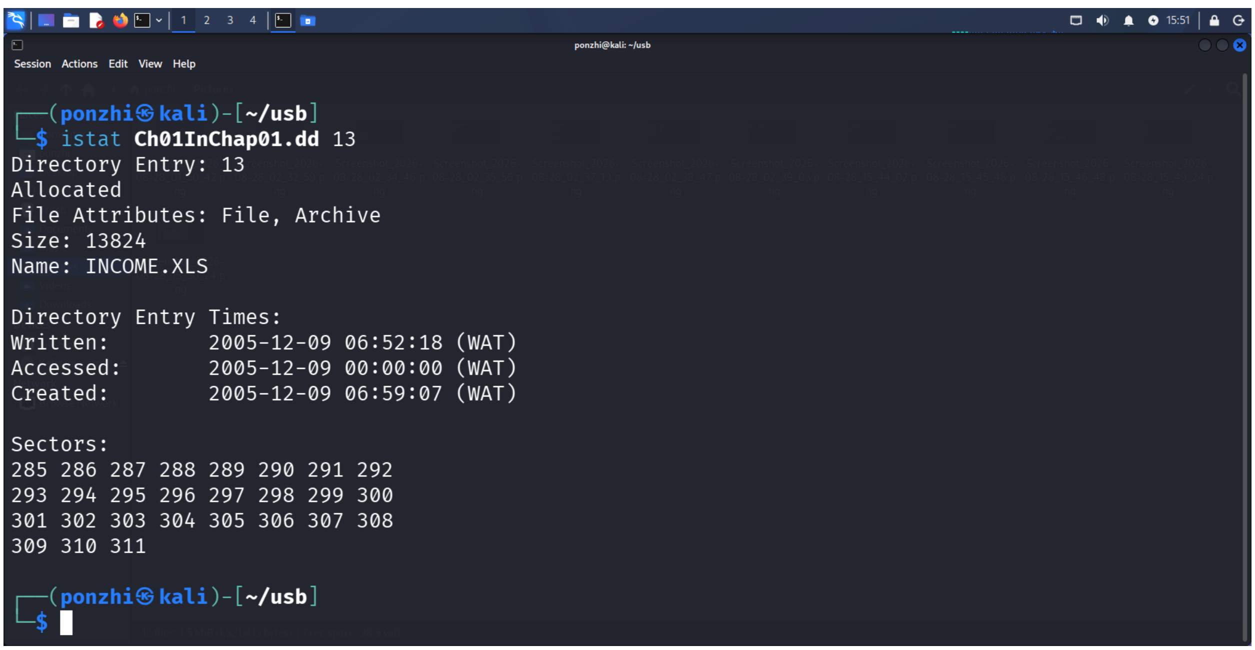
List the details about units and can extract the unallocated space of the file system.



Extract the contents of a given data unit in hex



Showing a file with multiple sectors

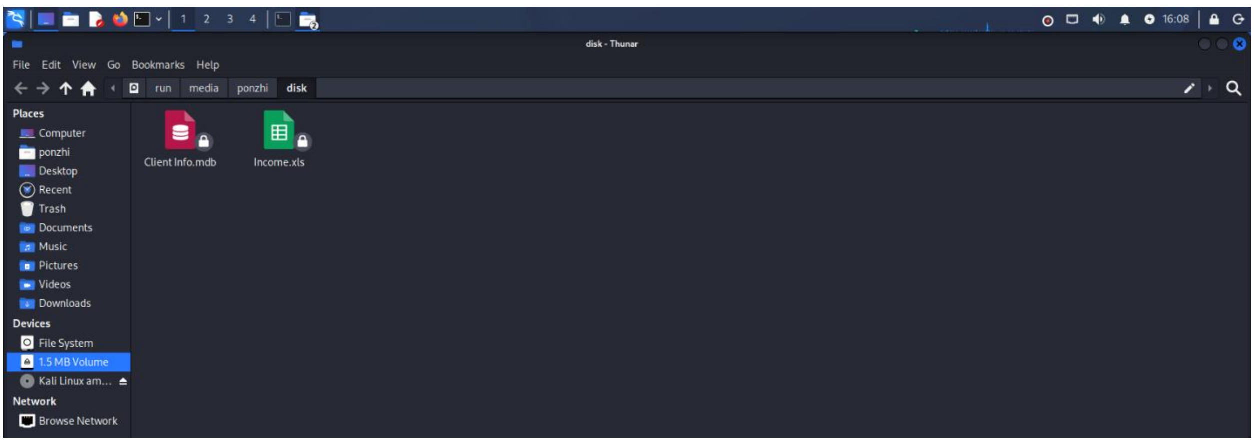


Mounting a USB image

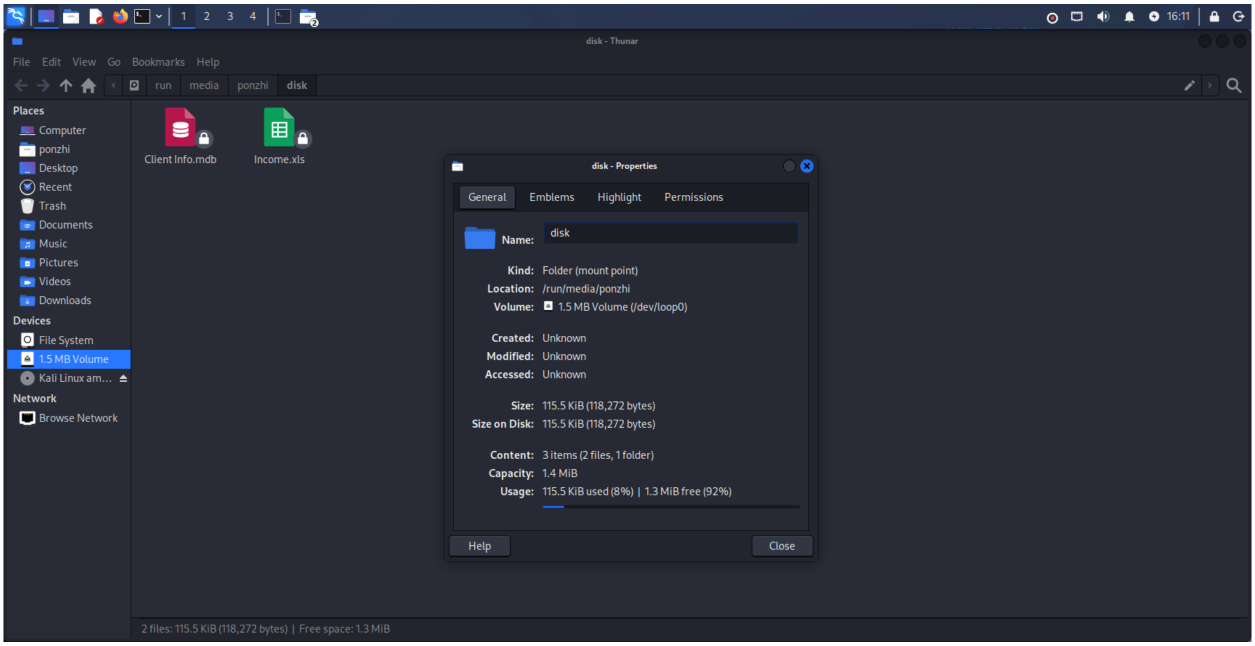
Partscan: This option tells losetup to scan the partition table in the specified disk image (Ch01InChap01.dd) and automatically create loop devices for each partition found. This is useful when dealing with disk images that contain multiple partitions.

Find: This option instructs losetup to find the first available loop device. When combined with --show, it ensures that the command displays the path of the loop device that is being used

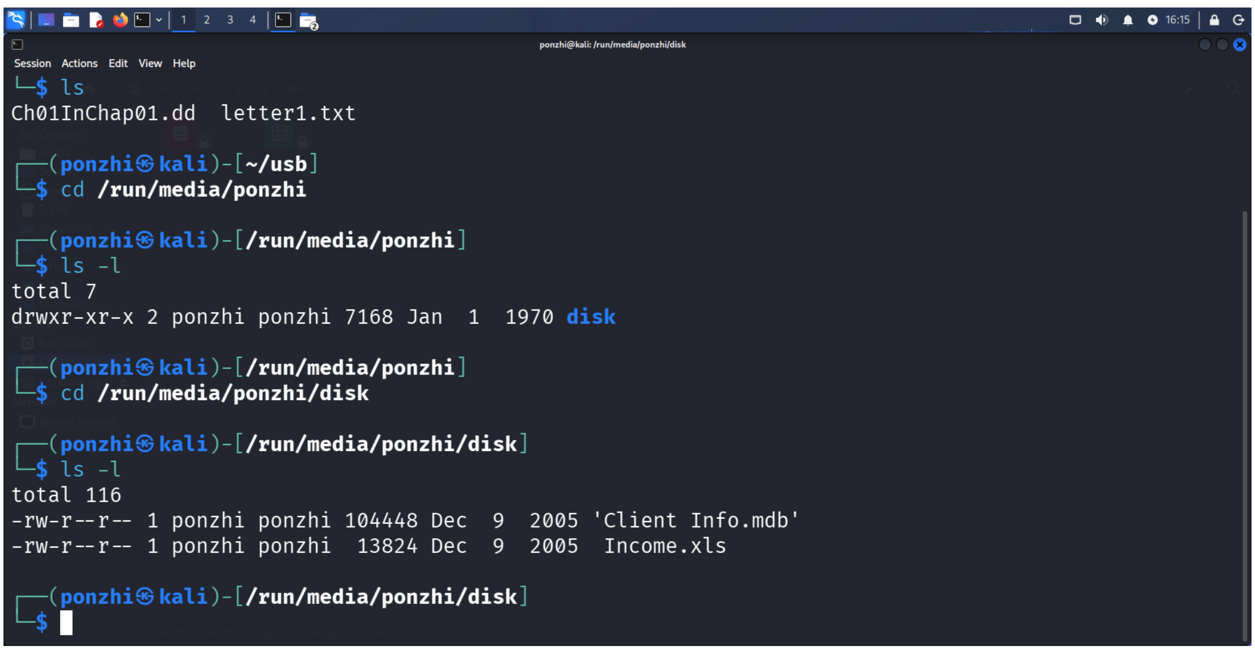
Showing Content



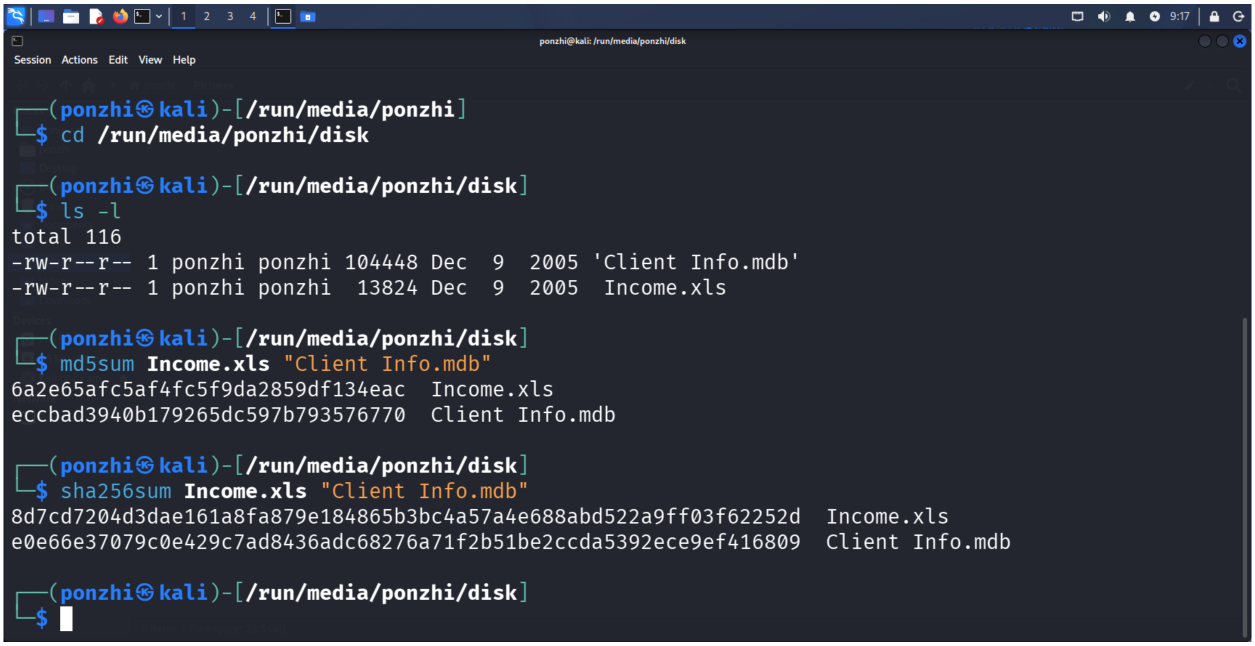
Showing the mounted Path



Showing the disk content



Hashing the files content



Conclusion

This examination successfully showed how to recover a deleted file from a forensic disk image using both graphical (Autopsy) and command-line (The Sleuth Kit) methods. We found the target file, INCOME.XLS, in the deleted files section of the Ch01InChap01.dd image using Autopsy's Deleted Files feature. We confirmed this finding through Sleuth Kit's fls command, which also showed the file’s metadata and inode address.

We tried three different methods to recover the file: icat and blkcat. After recovery, we opened the spreadsheet in Excel, which confirmed that it was complete and readable. This step proved that we recovered not just raw data but usable information.

Throughout this analysis, we ensured the original evidence image remained unchanged. We recorded the MD5 and SHA-256 hash values for Ch01InChap01.dd before starting our work, and they stayed the same, following proper procedures to maintain the chain of custody.

This exercise highlighted the importance of checking results with different forensic tools and confirming that recovered evidence is both intact and practical to use, beyond just matching hash values.